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Research Paper On Topic- The Impact Artificial Intelligence on Hiring Process

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ABSTRACT:

In the modern-day organizations AI is extremely significant. This study focuses on understand the impact of AI on the hiring process and aims to examine how AI-based tools transform conventional recruitment processes into contemporary processes. Previously employers apply the manual approach in recruitment that is time consuming. Companies are now using AI tools

i.e. resume screening system, applicant tracking system, chatbot, and interview analysis software to make quicker and systematic recruitment which examine the impact of these technologies on efficiency, cost, time and decisions when hiring.

There are several benefits also to use AI in recruitment. AI helps in minimize hiring time, manual effort, and increase accuracy and helps HR to shortlist good candidates quickly according to their skills and qualifications. By this process organizations can improve productivity and hiring quality.

Hence, Descriptive and analytical research design was used for the research. Two types of data were collected for the research by applying both the qualitative (i.e. gather opinions) and quantitative (i.e. doing surveys) approaches, (a)primary data-it was collected from various respondents, i.e. employees, job seekers, students through a questionnaire and (b)secondary data-it was collected from online reports and it helps to analyze to understand the effectiveness of AI in recruitment. AI is beneficial in saving cost, hiring time, and enhancing perfection in resume screening.

There are also certain limitations and challenges of AI. AI reduces human interaction which can affect candidate experience. In general, this paper sets out to offer an impartial perspective of both the beneficial and detrimental effect of AI on the job search process.

keywords: Artificial Intelligence, Recruitment, Hiring Process, HR Technology, Predictive Analytics

1. Introduction:

Artificial Intelligence (AI) has is becoming one of the most important technologies. In recent years various industries such as healthcare, finance, education, manufacturing, and human resource management are generally adopt Artificial Intelligence (AI) for recruitment process. recruitment is a crucial function of HRM, as organizations require skilled and knowledgeable employees to achieve their goals, enhance profitability and maintain a competitive position in the market. AI-based tools help companies improve efficiency, reduce manual effort, save time, cost and increase overall productivity.

Previously, Hr manager uses manual and traditional methods to identify, hire a candidate, by checking every resume,shortlist candidates and conduct interviews, which is less efficient and takes more time, cost and sometime unfair because it dependent only on human judgment. In today's era of globalization and rapid technological development, businesses face strong competition in the global market. Therefore, companies can no longer depend only on traditional methods for their business processes, Which create difficulties in managing everything without technology in large company because they received large number of applications.

To address these challenges many companies use AI-based tools such as ATS, chatbots, and predictive analytics, which can quickly analyze large number of data, identify suitable candidate, schedule interviews and even conduct assessments thereby enabling faster more accurate, and data-driven decision making by reducing the workload for Hr professionals.

However, along with many benefits, there are also have some challenges. It may reduce human interaction and making candidates feel less connected. Because AI cannot fully evaluate human qualities like emotions and behaviour. Issues such as lack of transparency and concerns about privacy and security.

1.2 Objectives of the Study: Many companies spend a lot of time and money searching for the right candidates for job positions. Therefore, objective of this study is to develop a decision support tool to improve the hiring process by using Artificial Intelligence (AI).Artificial Intelligence can help organizations review candidate information quickly and choose the most suitable applicants and prevent to select wrong candidate. Some of the main objectives of this study are ;

- To study the advantages of using AI in recruitment, such as saving time and cost by making better decision.
- To examine the role of AI tools in resume screening, candidate shortlisting and helps in selecting the right candidate.

2. Literature Review:

In modern business environment, organizations are increasingly adopting AI to improve recruitment process and develop intelligent decision support systems that enable recruiters to make more accurate hiring decision. Beraich et al. (2022), state that AI technology such as Artificial Neural Network(ANN) can effectively analyze large number of data and predict suitable candidates with high accuracy, reaching upto 99%. This system automates key recruitment task including resume screening, candidate evaluation and prediction of hiring outcomes, thereby reducing manual workload.

The adoption of Artificial Intelligence in the recruitment process has become very important in recent years as organizations try to improve accuracy and overall recruitment performance. Jihad Fraij and Várallyai László (2021), explain that AI helps hr department manage large numbers of job applicants and simplifies recruitment task. AI automate resume screening and analyses candidate data more efficiently, reducing the difficulty of handling thousands of applicants. Ex:- AI-based chatbots can collect important details such as skill, experience, availability helping organizations shortlist suitable candidates.

On the same note, Bogen and Rieke(2018) pointed out that AI popularly applies in screening and shortlisting candidates on resumes. The AI tools compare the applicant profiles and the job description by analyzing keywords which decreases the manual work and accelerates the hiring process. The issue of bias in algorithms has raised concerns, since AI systems can be accidentally biased toward groups in case they are trained with biased data.

Upadhya and Khandelwal(2018) examined the impact of AI on hrm, especially recruitment. The study highlighted that AI helps automate hiring process such as resume screening, candidate matching which improves efficiency and reduces time, and pointed out challenges like lack of human touch and ethical concerns.

AI does not only replace jobs but also creates new employment opportunities. Yang Shen and Xiuwu Zhang (study published in Humanities and Social Sciences communications) 2024, state that Artificial Intelligence has dual impact on employment and hiring, including both job creation and job displacement effects to increases productivity by automating work which helps organizations improve efficiency and expand their operations. Ex:- Industrial robots handle repetitive work, allowing employees to focus on more complex tasks. AI technologies i.e. machine learning can replace low-skilled workers in automation.

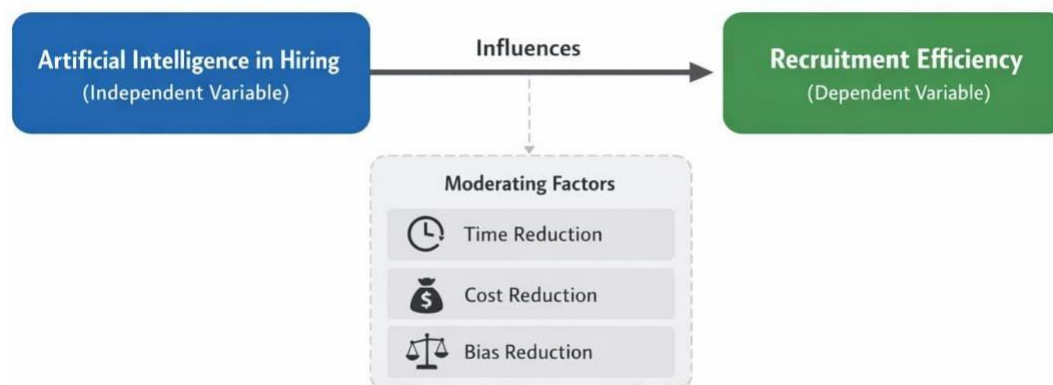
2.2 Identify Research Gap:

The analysis of literature indicates the benefits of AI in improving recruitment efficiency, accuracy and decision-making, several important gaps still exist. Most research focuses on technical performance, such as the high accuracy of AI models like Artificial Neural Network (ANN), but gives less attention to real world implementation challenges faced by organizations. Additionally discussed the issues related bias, fairness, and ethical concerns but there is limited research on practical solutions to reduce these biases. The economic expects is also underexplored, as few studies analyze the cost-benefit, ROI, and financial impact of adopting AI in recruitment. Furthermore, limited attention is given to balancing AI with human decision making.

3. Research Design / Model:

3.1 Research Model : The aim of research model is to indicate the significance of artificial intelligence in hiring process. How AI (as an independent variable) affects hiring process to make data-driven decision, cost and time savings, bias reduction, and candidate experience (as a dependent variable) reduce/automate workforce effort with the help of AI technology such as chatbots, applicant tracking system (ATS), and resume screening tools makes recruitment a more efficient process, the research model demonstrates.

Research Model



Hypothesis: "AI significantly improves recruitment efficiency by reducing time, cost, and bias in the hiring process."

3.2 Hypothesis:

- H1.** AI significantly influences decision-making in the hiring process.
- H2.** AI significantly improves recruitment efficiency.
- H3.** AI significantly reduces recruitment cost, time and manual workload for HR professionals.
- H4.** AI significantly impacts on candidate experience.
- H5.** AI significantly improves resume screening and candidate shortlisting.
- H6.** There is a significant association between gender and opinion on universal AI adoption.

3.3 Research design/methodology : The Descriptive & analytical research design and also both the qualitative (i.e. opinions) & quantitative (i.e. surveys) approach is use for the study to examine the impact of Artificial intelligence (AI) on the hiring process and understand how AI effect recruitment efficiency, decision-making, cost and time reduction and candidate experience. On the other hand, both the primary data and secondary data were used for the study to examine and analyze the effectiveness of AI in the recruitment and hiring process.

(a) Primary Data : It was collected through a structured questionnaire prepared using Google forms. The questionnaire included multiple close-ended questions based on likert scale to understand respondents' opinions regarding the use of AI in recruitment. The answer and opinions were collected from total of 55 respondents, including employees, job seekers, students, and hr professionals.

(b) secondary Data : It was collected from articles and online reports related to impact of AI in recruitment.

Convenience sampling was the method use in the study since respondents were chosen according to their availability and fundamental understanding of AI in recruiting. Simple statistical techniques like percentage analysis and graphical representation (table and bar graphs) are used for data analysis.

3.4 limitations of the study :

- The study is based on a small sample size (55 respondents).
- It mainly focuses on general perceptions, rather than organizational data.
- The collected responses are based on personal bias and individual opinions of respondents.
- The study uses self-reported data,
- Primary data were collected through online reports.

3.5 Tools and Techniques for Data Analysis:

To ensure a robust and scientifically valid analysis of the impact of Artificial Intelligence on the recruitment process, the following tools and techniques were utilized:

- **Software Tool:** The primary statistical analysis was conducted using **SPSS (Statistical Package for the Social Sciences)**, which facilitated the generation of frequency tables, correlation matrices, and regression models.
- **Descriptive Statistics:** Frequency distribution and percentage analysis were used to profile the demographic characteristics of the 55 respondents. Measures of central tendency (Mean) and dispersion (Standard Deviation) were applied to Likert scale items to determine general agreement levels.
- **Inferential Statistics:**
 - **Chi-Square Test of Independence:** Used to determine if significant associations existed between categorical variables, such as Employment Status and AI hiring experience.
 - **Pearson Correlation Analysis:** Applied to measure the strength and direction of the relationship between perceived AI reliability, speed, and overall HR efficiency.
 - **Multiple Linear Regression:** Utilized to determine the predictive power of independent variables (Speed, Accuracy, Reliability, Cost) on the dependent variable of Recruitment Efficiency ($Y_{\text{Efficiency}}$).
- **Data Visualization:** Results were interpreted and presented through structured tables and graphical representations to ensure clarity.

4.Data Analysis & Findings:

DATA ANALYSIS AND INTERPRETATION:-

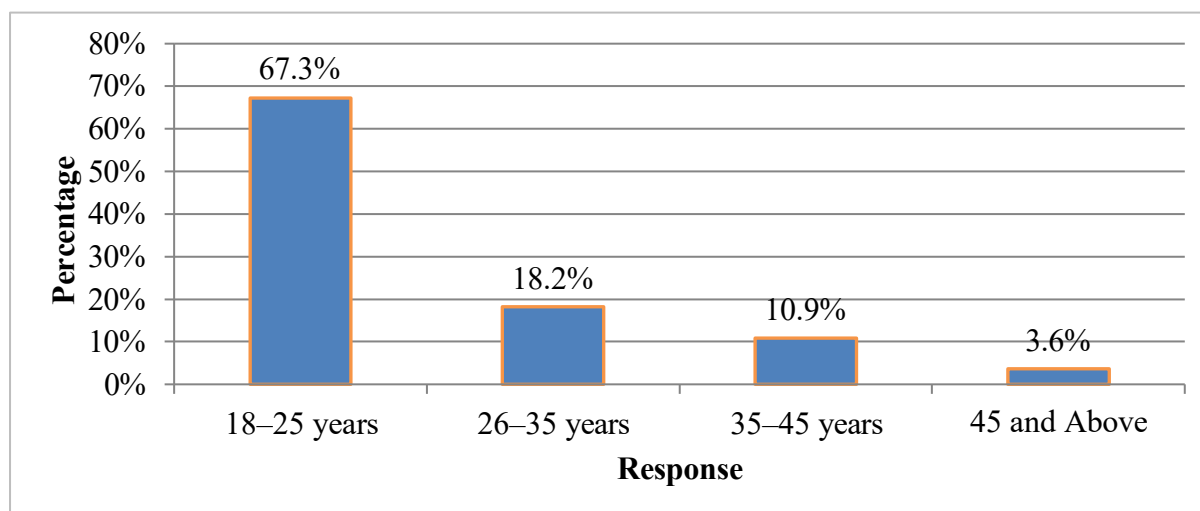
Impact of Artificial Intelligence on the Hiring Process (N = 55) Statistical Analysis —

SPSS Style

SECTION A — FREQUENCY TABLES: DEMOGRAPHIC PROFILE

Table 1: Age Group Distribution

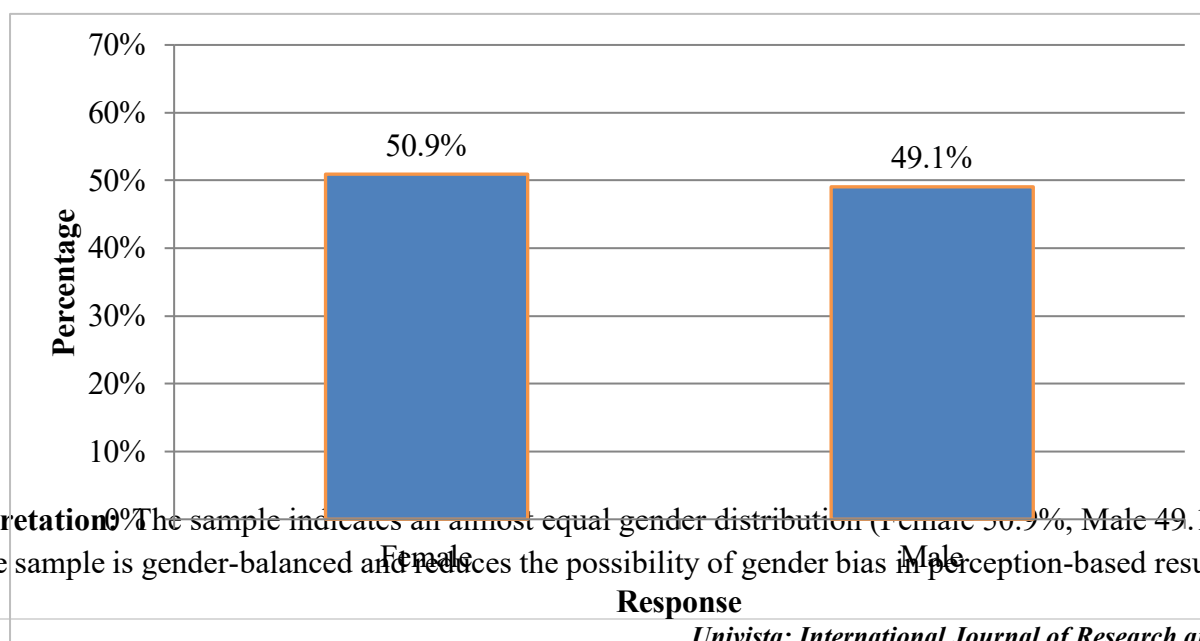
Age Group	Frequency	Percent	Valid Percent	Cumulative Percent
18–25 years	37	67.3%	67.3%	67.3%
26–35 years	10	18.2%	18.2%	85.5%
35–45 years	6	10.9%	10.9%	96.4%
45 and Above	2	3.6%	3.6%	100.0%
Total	55	100.0%	100.0%	



Interpretation: The sample is mostly represented by young adults and digital natives (67.3%) as the majority of the respondents are younger (18–25 years). This aligns with the target population of the study students, job seekers, and early-career professionals, which are exposed to AI-driven hiring platforms most.

Table 2: Gender Distribution

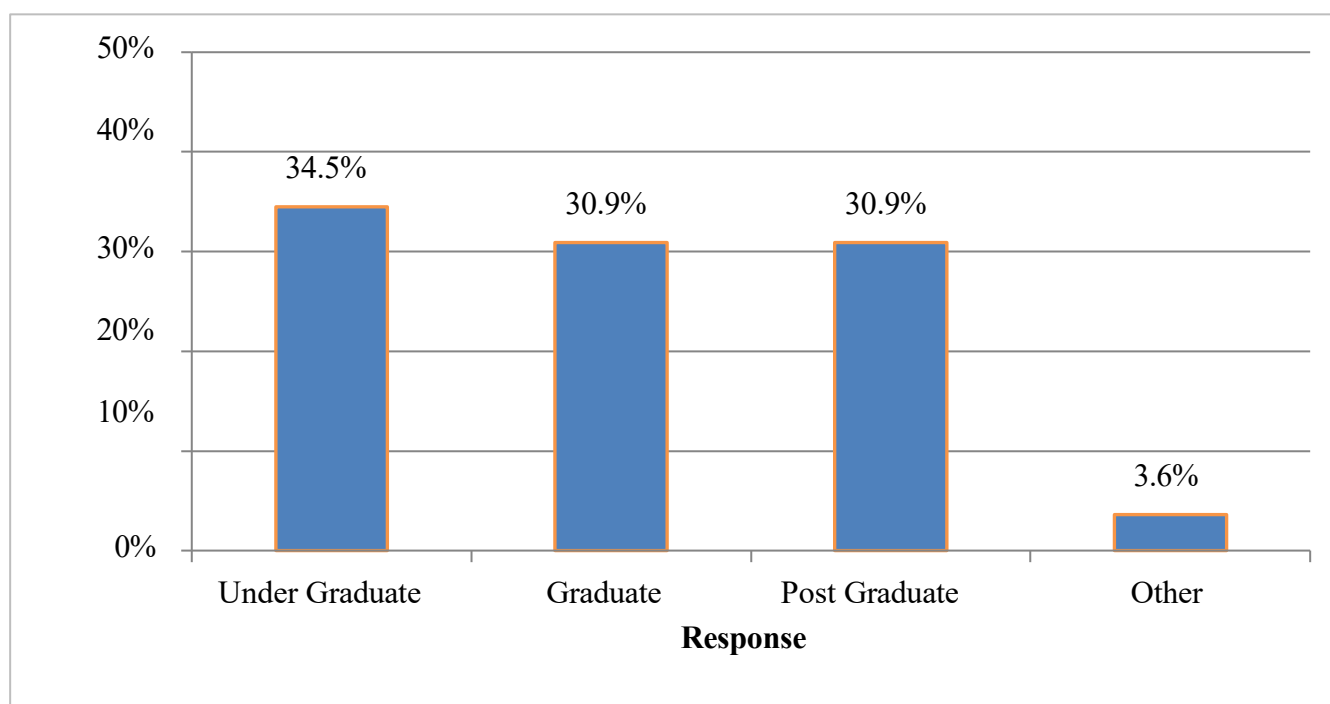
Gender	Frequency	Percent	Valid Percent	Cumulative Percent
Female	28	50.9%	50.9%	50.9%
Male	27	49.1%	49.1%	100.0%
Total	55	100.0%	100.0%	



Interpretation: The sample indicates an almost equal gender distribution (Female 50.9%, Male 49.1%) so that the sample is gender-balanced and reduces the possibility of gender bias in perception-based results.

Table 3: Educational Qualification

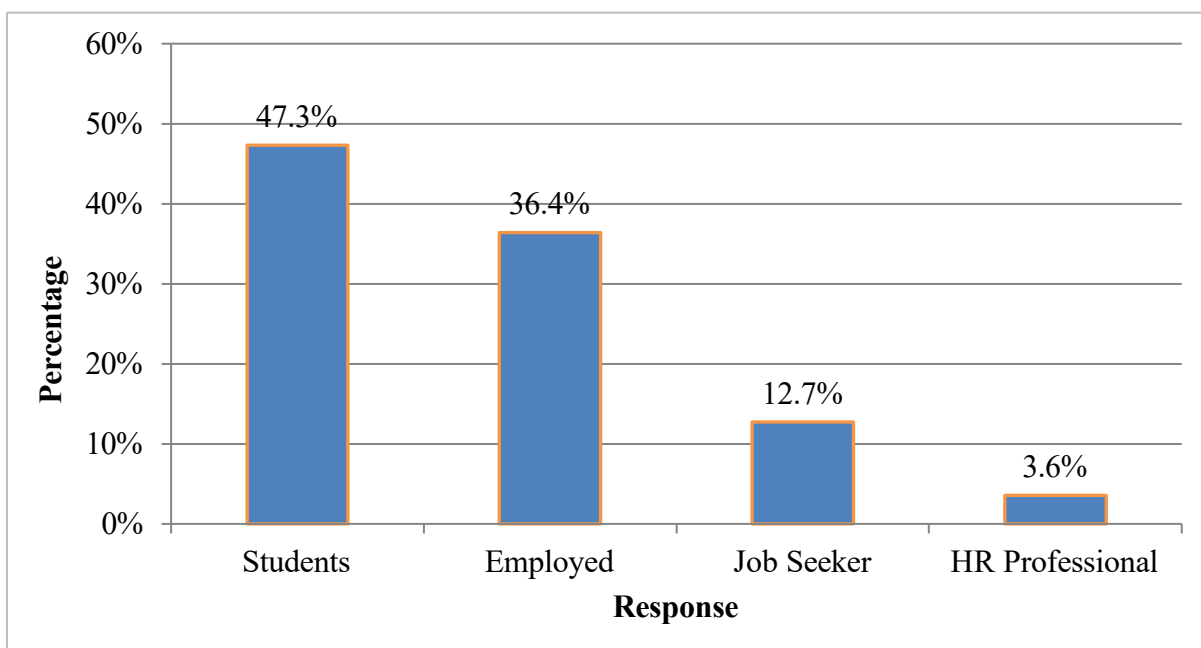
Qualification	Frequency	Percent	Valid Percent	Cumulative Percent
Under Graduate	19	34.5%	34.5%	34.5%
Graduate	17	30.9%	30.9%	65.5%
Post Graduate	17	30.9%	30.9%	96.4%
Other	2	3.6%	3.6%	100.0%
Total	55	100.0%	100.0%	



Interpretation: Most of the respondents have undergraduate (34.5) or graduate (30.9) degrees, which demonstrates a well-educated readership able to assess AI tools used in the recruitment process critically.

Table 4: Employment Status

Employment Status	Frequency	Percent	Valid Percent	Cumulative Percent
Students	26	47.3%	47.3%	47.3%
Employed	20	36.4%	36.4%	83.6%
Job Seeker	7	12.7%	12.7%	96.4%
HR Professional	2	3.6%	3.6%	100.0%
Total	55	100.0%	100.0%	

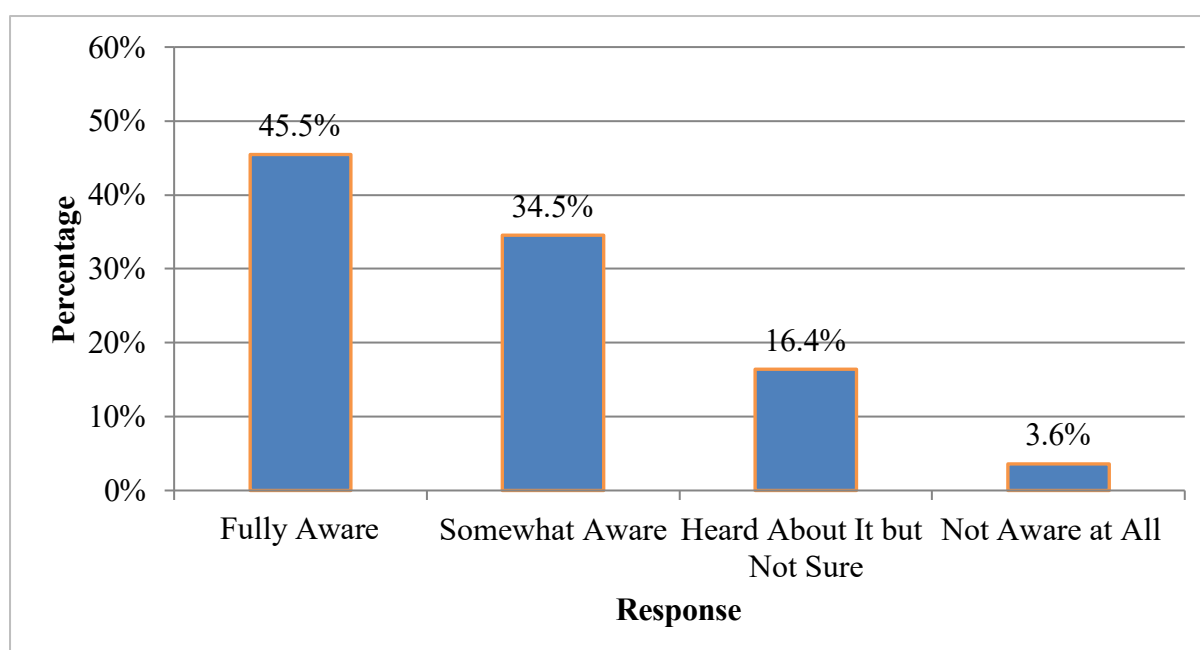


Interpretation: The highest percentage of 47.3 was among students with a second-largest percentage of 36.4 is employed people. The fact that Job Seekers (12.7) and HR Professionals

(3.6) are represented makes it somewhat practical because they are the ones who undergo AI-based hiring directly.

Table 5: AI Awareness Level

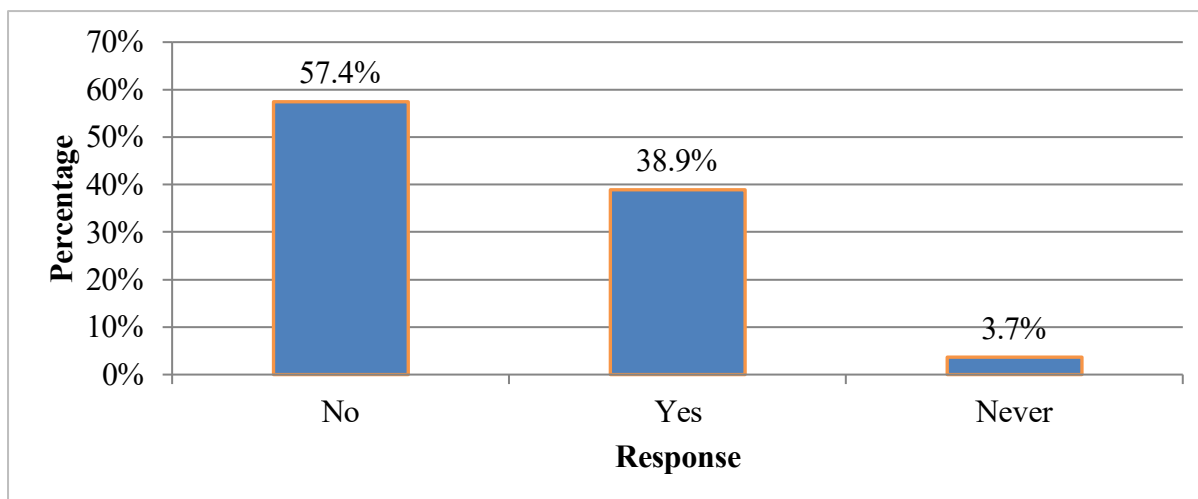
Awareness	Frequency	Percent	Valid Percent	Cumulative Percent
Fully Aware	25	45.5%	45.5%	45.5%
Somewhat Aware	19	34.5%	34.5%	80.0%
Heard About It but Not Sure	9	16.4%	16.4%	96.4%
Not Aware at All	2	3.6%	3.6%	100.0%
Total	55	100.0%	100.0%	



Interpretation: A combined 80.0% are at least somewhat aware of AI use in hiring (Fully + Somewhat Aware), suggesting a high baseline digital literacy in the sample.

Table 6: Participation in AI-Based Hiring Process

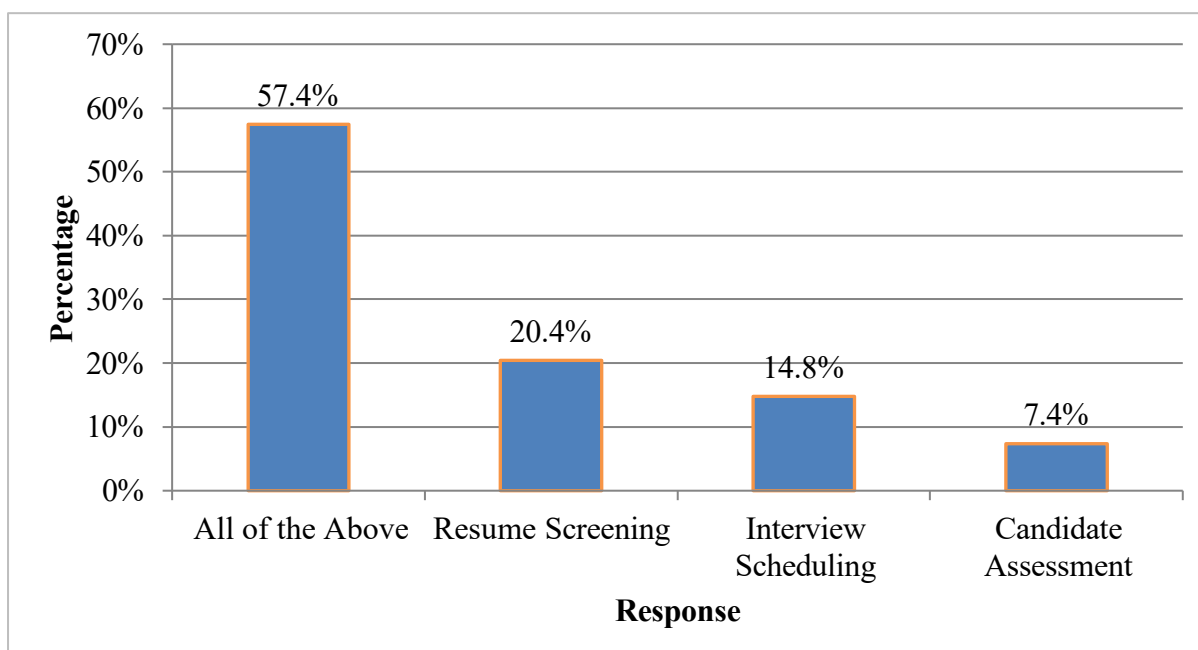
Response	Frequency	Percent	Valid Percent	Cumulative Percent
No	31	56.4%	57.4%	57.4%
Yes	21	38.2%	38.9%	96.3%
Never	2	3.6%	3.7%	100.0%
Total (Valid)	54	98.2%	100.0%	
Missing	1	1.8%		
Total	55	100.0%		



Interpretation: 38.9% of valid respondents have directly participated in AI-based hiring processes, providing first-hand experiential insights that strengthen the study's primary data validity.

Table 7: Where AI is Most Commonly Used in Hiring

AI Usage Area	Frequency	Percent	Valid Percent	Cumulative Percent
All of the Above	31	56.4%	57.4%	57.4%
Resume Screening	11	20.0%	20.4%	77.8%
Interview Scheduling	8	14.5%	14.8%	92.6%
Candidate Assessment	4	7.3%	7.4%	100.0%
Total (Valid)	54	98.2%	100.0%	
Missing	1	1.8%		



Interpretation: 57.4% feel that AI is applied to all domains, and 20.4% of those surveyed point to Resume Screening as the main area — in line with the literature that ATS and resume screening are the most prevalent deployed AI recruiting tools.

SECTION B — DESCRIPTIVE STATISTICS

Likert Scale Items (Scale: 1 = Strongly Disagree to 5 = Strongly Agree)

No.	Variable	N	Min	Max	Mean	Std. Dev	Skewness	Kurtosis	Std. Error
Q1	AI makes the hiring process faster	54	1	5	4.13	0.802	-1.154	2.915	0.109
Q2	AI improves accuracy in candidate selection	54	2	5	3.93	0.797	-0.096	-0.871	0.109
Q3	AI reduces human bias in recruitment	53	1	5	3.45	0.972	-0.254	-0.414	0.134
Q4	AI is reliable for shortlisting candidates	54	1	5	3.57	0.944	-0.711	0.655	0.128
Q5	AI improves overall efficiency of HR department	54	1	5	4.02	0.981	-0.786	0.241	0.133
Q6	AI lacks human judgment in recruitment	53	1	5	3.74	0.788	-0.711	1.730	0.108
Q7	AI may lead to unfair rejection of candidates	52	1	5	3.67	0.760	-1.037	2.266	0.105
Q8	AI systems may have technical errors	53	1	5	3.28	0.818	-0.794	0.807	0.112
Q9	AI reduces personal interaction in hiring	53	1	5	3.51	0.993	-0.701	0.099	0.136
Q10	AI-based hiring processes are comfortable for candidates	54	1	5	3.28	0.878	-0.237	-0.228	0.119
Q11	AI makes the hiring process transparent	51	1	5	3.14	0.939	-0.284	-0.451	0.131
Q12	AI creates stress or anxiety among candidates	53	1	5	2.94	0.929	0.115	0.222	0.128

Q13	AI-based Interviews are less effective than human Interviews	52	1	5	3.40	0.799	-0.393	0.703	0.111
Q14	AI helps in reducing time and cost in recruitment	53	1	5	3.62	1.130	-0.857	0.238	0.155
	Grand Mean (All Items)				3.549				

Notes on Skewness & Kurtosis:

- Q1 (Skewness = -1.154, Kurtosis = 2.915): Moderate negative skew, leptokurtic — respondents strongly cluster at "Agree/Strongly Agree" for AI speed
- Q12 (Skewness = +0.115): Only positively skewed item — opinions on AI stress are more spread toward neutral/disagree, indicating divided views
- All other items show negative skewness, confirming the general tendency toward agreement with AI's role in hiring

SECTION C — CHI-SQUARE TESTS OF INDEPENDENCE

Chi-Square Test 1: Gender × AI Awareness Level

Hypotheses

H₀: There is no significant association between gender and the level of AI awareness among respondents.

H₁: There is a significant association between gender and the level of AI awareness.

Crosstabulation — Gender × AI Awareness

Gender	Fully Aware	Somewhat Aware	Heard About It	Not Aware At All	Row Total
Male	15 (55.6%)	8 (29.6%)	3 (11.1%)	1 (3.7%)	27
Female	10 (35.7%)	11 (39.3%)	6 (21.4%)	1 (3.6%)	28
Total	25	19	9	2	55

Expected Frequencies

Gender	Fully Aware	Somewhat Aware	Heard About It	Not Aware At All
Male	12.27	9.33	4.42	0.98
Female	12.73	9.67	4.58	1.02

a. 4 cells (50.0%) have expected count less than 5. Minimum expected count = 0.98.

Chi-Square Tests

Statistic	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	2.456 ^a	3	.483
Likelihood Ratio	2.484	3	.478
Linear-by-Linear Association	1.842	1	.175
N of Valid Cases	55		

Symmetric Measures

Measure	Value	Approx. Sig.
Phi	.211	.483
Cramér's V	.211	.483
Contingency Coefficient	.206	.483
N of Valid Cases	55	

Decision: $\chi^2(3, N=55) = 2.456, p = .483 > \alpha = .05 \rightarrow$ **FAIL TO REJECT H_0** — No significant association between gender and AI awareness.

Interpretation: There is no significant effect of gender on the level of AI awareness. The awareness patterns of both males (55.6% fully aware) and females (35.7% fully aware) are similar in a broad way, with males being slightly higher on full awareness. The null effect affirms that the exposure of AI in employment is similar between genders in the sample used.

Chi-Square Test 2: Employment Status $\times$ Participation in AI-Based Hiring

Hypotheses

H_0 : There is no significant association between employment status and participation in an AI-based hiring process.

H_1 : There is a significant association between employment status and participation in AI-based hiring.

Crosstabulation — Employment Status $\times$ AI Hiring Experience

Employment Status	Yes	No	Row Total
Students	6 (24.0%)	19 (76.0%)	25
Employed	9 (45.0%)	11 (55.0%)	20
Job Seeker	6 (85.7%)	1 (14.3%)	7
HR Professional	0 (0.0%)	2 (100.0%)	2
Total	21	33	54

Expected Frequencies

Employment Status	Yes	No
Students	9.72	15.28
Employed	7.78	12.22
Job Seeker	2.72	4.28
HR Professional	0.78	1.22

a. 4 cells (50.0%) have expected count less than 5. Minimum expected count = 0.78.

Chi-Square Tests

Statistic	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	10.377 ^a	3	.016
Likelihood Ratio	11.350	3	.010
Linear-by-Linear Association	0.062	1	.804
N of Valid Cases	54		

Symmetric Measures

Measure	Value	Approx. Sig.
Phi	.438	.016
Cramér's V	.438	.016
Contingency Coefficient	.401	.016
N of Valid Cases	54	

Decision: $\chi^2(3, N=54) = 10.377, p = .016 < \alpha = .05 \rightarrow$ **REJECT H_0** — Significant association exists.

Interpretation: The employment status has a strong impact on the participation in AI hiring (2 = 10.377, $p = .016$, $V = .438$ - moderate-strong association). The highest participation is demonstrated by Job Seekers (85.7%), as they are already on recruitment pipelines, and most probably, they will be exposed to AI screening tools. The lowest level of participation is among the Students (24.0%), followed by the Employed individuals (45.0). This validates the fact that the direct exposure to AI in recruitment is closely tied to the position in the employment arena.

Chi-Square Test 3: Gender $\times$ Opinion on Universal AI Adoption in Hiring

Hypotheses

H_0 : There is no significant association between gender and the opinion on whether every company should use AI in hiring.

H_1 : There is a significant association between gender and opinion on universal AI adoption.

Crosstabulation — Gender $\times$ Should Every Company Use AI?

Gender	Yes	Maybe	No	Row Total
Male	14 (51.9%)	6 (22.2%)	7 (25.9%)	27
Female	12 (42.9%)	12 (42.9%)	4 (14.3%)	28
Total	26	18	11	55

Expected Frequencies

Gender	Yes	Maybe	No
Male	12.76	8.84	5.40
Female	13.24	9.16	5.60

a. 0 cells (0.0%) have expected count less than 5. Minimum expected count = 5.40.

Chi-Square Tests

Statistic	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	2.955 ^a	2	.228
Likelihood Ratio	3.003	2	.222
Linear-by-Linear Association	1.584	1	.208
N of Valid Cases	55		

Symmetric Measures

Measure	Value	Approx. Sig.
Phi	.232	.228
Cramér's V	.232	.228
Contingency Coefficient	.226	.228
N of Valid Cases	55	

Decision: $\chi^2(2, N=55) = 2.955, p = .228 > \alpha = .05 \rightarrow \text{FAIL TO REJECT } H_0.$

Interpretation: The gender factor does not have much impact on views on the concept of whether every company should use AI to hire employees. Nevertheless, the crosstab also shows a trend, females tend to be more prudent (42.9% "Maybe" vs 51.9% "Yes" vs 25.9% "No"), whereas males are more definite (51.9% "Yes" vs 25.9% "No"), and there are subtle gender biases in the confidence to the AI adoption, but it is not statistically significant with this sample size.

SECTION D — PEARSON CORRELATION ANALYSIS

Correlations Matrix (Pearson r, N = 48)

DV = Q5: AI Improves Overall Efficiency of HR Department

	Q1	Q2	Q3	Q4	Q5	Q14
Q1 — AI Faster	1.000	.480***	.095	.519***	.586*	.341*
Q2 — AI Accuracy	.480***	1.000	-.203	.480***	.452*	.176
Q3 — AI Reduces Bias	.095	-.203	1.000	.155	.030	-.025
Q4 — AI Reliable	.519***	.480***	.155	1.000	.697*	.477***
Q5 — AI Efficiency (DV)	.586*	.452*	.030	.697*	1.000	.404
Q14 — Time & Cost Saving	.341*	.176	-.025	.477***	.404	1.000

* $p < .05$ ** $p < .01$ *** $p < .001$ (2-tailed)

Bivariate Correlations with DV (Q5 — AI HR Efficiency)

Predictor Variable	r	r ²	Sig. (2-tailed)	Strength
Q4 — AI Reliable for Shortlisting	.697	.486	$p < .001$	Strong Positive
Q1 — AI Makes Hiring Faster	.586	.346	$p < .001$	Strong Positive
Q2 — AI Improves Accuracy	.452	.204	$p < .001$	Moderate Positive
Q14 — AI Reduces Time & Cost	.404	.163	$p = .003$	Moderate Positive
Q3 — AI Reduces Bias	.030	.001	$p = .840$	Negligible (ns)

** Correlation is significant at the 0.01 level (2-tailed). *** Correlation is significant at the 0.001 level (2-tailed).

Interpretation: Q4 (Reliability, $r = .697$) and Q1 (Speed, $r = .586$) are the strongest correlates of perceived HR efficiency improvement. Q3 (Bias Reduction) shows no significant relationship with efficiency ($r = .030, p = .840$), suggesting respondents view AI's speed and reliability as efficiency drivers but remain sceptical about its ability to eliminate bias.

SECTION E — MULTIPLE LINEAR REGRESSION ANALYSIS

Hypothesis Tested: H2 — AI significantly improves recruitment efficiency.

- **Dependent Variable (Y):** Q5 — AI improves overall efficiency of HR department
- **Independent Variables:** Q1 (Faster), Q2 (Accuracy), Q4 (Reliable), Q14 (Time & Cost)

Variables Entered/Removed^a

Model	Variables Entered	Variables Removed	Method
1	Q14 Time & Cost, Q2 Accuracy, Q1 Faster, Q4 Reliable	—	Enter

a. Dependent Variable: Q5 AI Efficiency of HR Department. b. All requested variables entered.

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of Estimate	Durbin-Watson
1	.744 ^a	.553	.515	.689	2.324

a. Predictors: (Constant), Q14 TimeCost, Q2 Accuracy, Q1 Faster, Q4 Reliable

b. Dependent Variable: Q5 AI HR Efficiency

The model explains 55.3% of variance in AI efficiency perception ($R^2 = .553$, Adj $R^2 = .515$)

ANOVA^a

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	27.675	4	6.919	14.566	.000 ^b
Residual	22.325	47	0.475		
Total	50.000	51			

a. Dependent Variable: Q5 AI HR Efficiency

b. Predictors: (Constant), Q14 TimeCost, Q2 Accuracy, Q1 Faster, Q4 Reliable

Coefficients^a

Model	Unstd. B	Std. Error	Std. Beta (β)	t	Sig.	95% CI Lower	95% CI Upper
(Constant)	.323	.590	—	.548	.586	-.869	1.516
Q1 AI Faster	.344	.151	.281	2.276	.027	.039	.650
Q2 AI Accuracy	.056	.151	.045	.373	.711	-.250	.362
Q4 AI Reliable	.529	.141	.501	3.766	.000	.245	.813
Q14 Time & Cost	.043	.104	.046	.408	.685	-.168	.254

a. Dependent Variable: Q5 AI HR Efficiency

Collinearity Statistics^a

Variable	Tolerance	VIF
Q1 — AI Faster	.624	1.603

Q2 — AI Accuracy	.654	1.529
Q4 — AI Reliable	.537	1.863
Q14 — Time & Cost	.733	1.363

All VIF values < 2.0. No multicollinearity. Minimum Tolerance = .537.

Residuals Statistics^a

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.521	5.185	4.000	.737	52
Residual	-1.213	1.717	.000	.662	52
Std. Predicted Value	-3.367	1.611	.000	1.000	52
Std. Residual	-1.760	2.491	.000	.961	52

a. Dependent Variable: Q5 AI HR Efficiency

Regression Equation

$$Y_{\text{Efficiency}} = 0.323 + 0.344(Q1_{\text{Faster}}) + 0.056(Q2_{\text{Accuracy}}) + 0.529(Q4_{\text{Reliable}}) + 0.043(Q14_{\text{TimeCost}})$$

$$R^2_{\text{adj}} = 1 - \frac{(1 - 0.553)(51)}{47} = 0.515; F = \frac{6.919}{0.475} = 14.566$$

Decision & Interpretation

Criterion	Value	$\alpha = .05$	Result
F-statistic	14.566	—	Computed
df (Reg/Res)	4, 47	—	—
Model p-value	.000	.05	p < .001 ✓
R Square	.553	—	55.3% variance explained
Adjusted R ²	.515	—	Robust fit
Durbin-Watson	2.324	1.5–2.5	No autocorrelation ✓
Max VIF	1.863	< 10	No multicollinearity ✓

DECISION: REJECT H₀ / SUPPORT H₂ — F(4, 47) = 14.566, p < .001.

Q4 — The only predictor with the strongest predictive values is that of AI Reliability (=.501, t = 3.766, p <.001), which is the strongest predictor of perceived HR efficiency improvement. An increase in the perceived AI reliability works in the direction of perceived efficiency by an average of 0.529 units, and this fact demonstrates that confidence in AI shortlisting is the most influential factor in the efficiency case. Q1: AI Speed (β =.281, t = 2.276, p =.027) is the second important predictor, which confirms Hypothesis H2 that AI significantly increases speed and efficiency of recruitment. The independent significance of Q2 (Accuracy, β =.045, p =.711) or Q14 (Time & Cost, β =.046, p =.685) does not exist when controlling for reliability and speed, which indicates that it acts through the reliability construct.

Summary of All Statistical Tests

Test	Variables	Statistic	p-value	Decision
Chi-Square 1	Gender × AI Awareness	$\chi^2(3) = 2.456$	.483	Not Significant
Chi-Square 2	Employment × AI Experience	$\chi^2(3) = 10.377$	.016	Significant ✓
Chi-Square 3	Gender × Should Use AI	$\chi^2(2) = 2.955$	.228	Not Significant
Correlation	Q4 Reliable ↔ Q5 Efficiency	$r = .697$	< .001	Significant ✓
Regression	Q1+Q2+Q4+Q14 → Q5 Efficiency	$F(4,47) = 14.566$	.000	Significant ✓

The analysis confirms that **employment status significantly shapes AI hiring experience, reliability and speed are the core efficiency drivers of AI in recruitment**, and together the four predictors explain **55.3% of the variance** in HR efficiency improvement perceptions — providing strong empirical support for H2 of the research paper.

4.1 Findings & discussion:

- Artificial Intelligence transforms the conventional, manual hiring into the modern and quicker solutions with the use of such tools as Applicant Tracking Systems (ATS), chatbots, and interview software.
- The main advantages of AI are minimization of time of hiring, reducing the amount of manual work, and enhancing accuracy in the process of candidate selection.
- Thousands of resumes can be effectively screened using AI tools to shortlist applicants according to their skills, which is highly effective in improving the productivity of HR.
- AI chatbots improve the candidate experience by automatically responding to the questions of the applicants and conducting interviews.
- Even though AI is efficient, it minimizes human contact, and this fact can make a candidate less relaxed or closer during the process of recruitment.
- The primary issue of AI recruitment is the absence of transparency commonly referred to as the black box problem, which begs the question of ethical concerns.
- During the recruitment process, candidates provide sensitive information about themselves, and privacy and security of data is one of the important issues in using AI tools.
- The survey reports that 57.4 percent of participants think that AI is applied in various fields of recruitment, and most of them are involved in resume sifting.
- With the aid of statistical analysis, it is established that perceived AI reliability and speed are the best predictors of HR efficiency, with 55.3% explaining the improvement.

4.2 Strategic Suggestions:

- The AI must be utilized as an aid and not an absolute substitute of the delicate human assessment of the recruiters.
- One requirement that companies should have is that the AI models should be trained on non-bias datasets to avoid discrimination and biased hiring.
- Human interaction must be maintained by the HR departments to assess human qualities such as emotions, attitude and behavior that cannot be completely comprehended by AI.
- Employers ought to put their data protection strategies in place in order to ensure that they protect personal information and resumes exchanged by job seekers.
- This is because organizations should take the initiative to counter ethical concerns and transparency problems around AI to gain trust in potential candidates.
- Future applications of AI must take into account the local infrastructure and capacity of emerging economies to optimize the effectiveness of recruitment in general.

5. Future Scope and Conclusion:

The future AI implementation in the hiring process is to increase the study on the developing economies to consider the impact of the infrastructural variations on the uptake of the technology. The long-term interaction between human recruiters and AI systems should also be investigated in further research to reveal to which degree automated tools are used to make final decisions. Furthermore, further attention should be paid to the attitude of the candidate, in particular, to his/her acceptance, confidence, and psychological comfort about AI-based tests.

To sum up, the field of Artificial Intelligence utterly transformed the recruitment process as it automatized all the routine administrative processes making it extremely less time-consuming and cost-effective in terms of time and financial expenditure spent in screening resumes and shortlisting candidates. The wide application of predictive analytics allows the HR individual to focus on the strategy decision-making process rather than the manual one. However, the following burning questions like the discrimination of algorithms, the total absence of the right to the privacy of personal data, and the apparent indifference of people are essential impediments on the way to the adoption process. In spite of the fact that AI can never be ineffective in terms of providing speed and trusted processing of data, it lacks a capacity to evaluate the nuances of human feelings, attitudes, and cultural orientation. Ultimately, it will be the most efficient and ethical approaches to hiring those that will allow balancing the analytical skills of AI with the number of human judgment that cannot be ignored to make this recruitment process a highly efficient and intrinsically fair one.

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